Phase 3 report

GameMapTest (7 tests)

These tests validate the correctness of the game's map system, including:

- Tile retrieval and out-of-bounds checking
- Walkable vs. non-walkable tile permissions
- Map loading consistency

This achieved **96% line coverage**, indicating that the map logic is thoroughly validated.

CellTest (5 tests)

Covers:

- Correct handling of cell types
- State transitions
- Interactions between map tiles and gameplay logic

PlayerTest (8+ tests)

Ensures:

- Player movement respects boundaries
- Direction and speed updates behave correctly
- Health and scoring updates trigger properly
- Edge-case scenarios (e.g., attempts to move outside the map)

EntityTest

Tests common entity attributes, including:

- Position and velocity setters
- Basic state changes
- Standard update behavior

CollectibleTest

Validates scoring item behavior:

- Pick-up logic
- Proper removal from the game world
- Score increment accuracy

TrapTest

Tests trap functionality:

- Trigger activation
- Damage application
- Cooldown logic preventing repeated instant activation

CursedIdolTest

Validates cursed idol mechanics:

- State transitions
- Behavior triggered when the player interacts with the idol
- Correct update propagation in the game state

PathfinderTest

Focuses on AI pathfinding:

- Shortest-path generation
- Path validity
- Behavior when paths are blocked

AI coverage reached **64%**, which is strong for non-deterministic movement logic.

GameSettingsTest (11 tests)

Covers the game's difficulty system:

- Difficulty multipliers
- Gameplay parameter scaling
- Validation of invalid or extreme settings
- Stability when switching between difficulty levels

DifficultyLevelTest (17 tests)

Validates difficulty enum behavior:

- Difficulty progression
- Boundary behavior
- Correct parameter attachment (speed, spawn rate, score scaling)
- Error handling for unexpected inputs

ExtendedEntityTest (8 tests)

Adds coverage for:

- More complex player state transitions
- Collision edge cases (entity vs. boundary)
- Movement prediction logic

These tests helped identify and fix several inconsistencies in movement and entity updates.

Test Coverage Analysis

JaCoCo was used to generate coverage reports. Some packages (UI, audio, JavaFX screens) cannot be tested using Maven/JUnit, so their coverage is intentionally 0%.

High-Coverage Areas

Package	Coverage	Status
domain.map	**96%**	Excellent
domain.entities	**89%**	Very Good
domain.collision	**84%**	Strong
domain.ai	**64%**	Moderate

These results show that the core game logic—movement, collisions, AI, items, and map behavior—is thoroughly tested.

Bugs Discovered and Fixed

HUD Score Not Updating

Fixed by ensuring UI observers receive score changes consistently.

Incorrect Package Declaration in GameView

Caused classpath errors; corrected to match directory structure.

Menu Background & Layout Scaling Problems

Resolved by adjusting layout constraints to ensure consistent appearance on all resolutions.

Difficulty Menu Scaling Issues

Fixed during UI review and menu restructuring.

Difficulty System Improvements

We expanded and corrected the difficulty system to ensure consistent gameplay scaling. Adjusted parameters include:

- Guardian speed
- Number of guardians

Testing revealed issues such as extreme speed scaling and incorrect spawn counts, which were fixed during Phase 3.